

Solid lipid nanocarriers to increase oral bioavailability of the peptide Leuprolide

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CONTEXT

Peptides are interesting therapeutic molecules with specific and potent action over a wide variety of disease. Their oral bioavailability is strongly limited by their **proteolytic degradation** in the duodenal lumen as well as their **poor permeability** across the intestinal epithelium.

In this study, we considered encapsulation of peptides in biodegradable and biocompatible Nanostructured Lipid Carriers (NLC) and Solid Lipid Nanoparticles (SLN) to overcome both oral bioavailability limiting aspects [1].

OBJECTIVES

- to develop a fabrication process for the encapsulation of Leuprolide (LEU) a hydrophilic model peptide in NLC and SLN
- to evaluate the capacity of the nanoparticles to protect LEU from proteolytic degradation
- to assess ability of the lipid-based nanosuspensions to increase LEU permeability across the intestinal barrier with Caco-2 cell model (enterocyte-like model) and Caco-2:HT29-MTX co-cultures (goblet cells, mucin-secreting cells).

METHODS

Hydrophobic Ion Pair (HIP) Formation

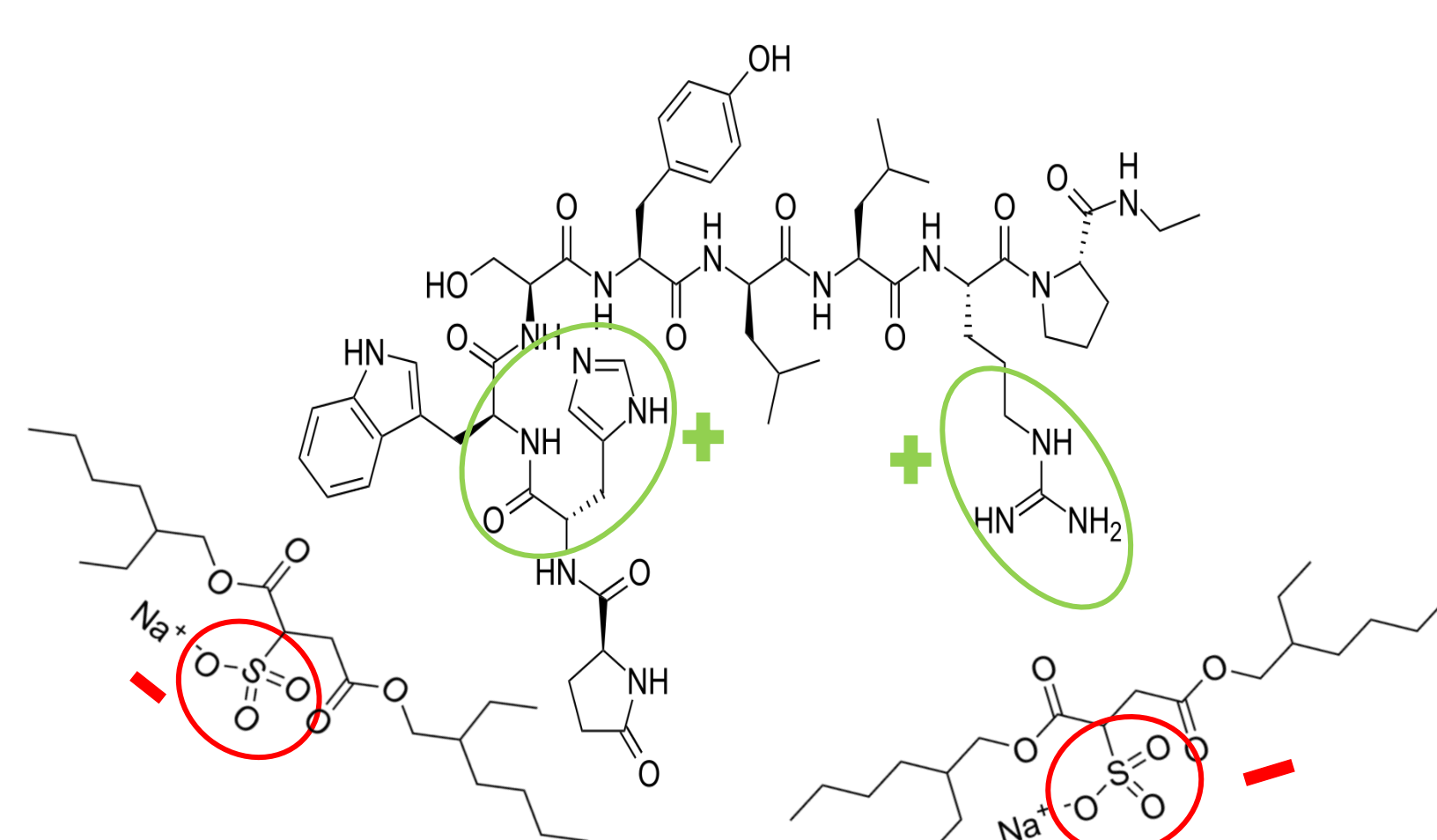
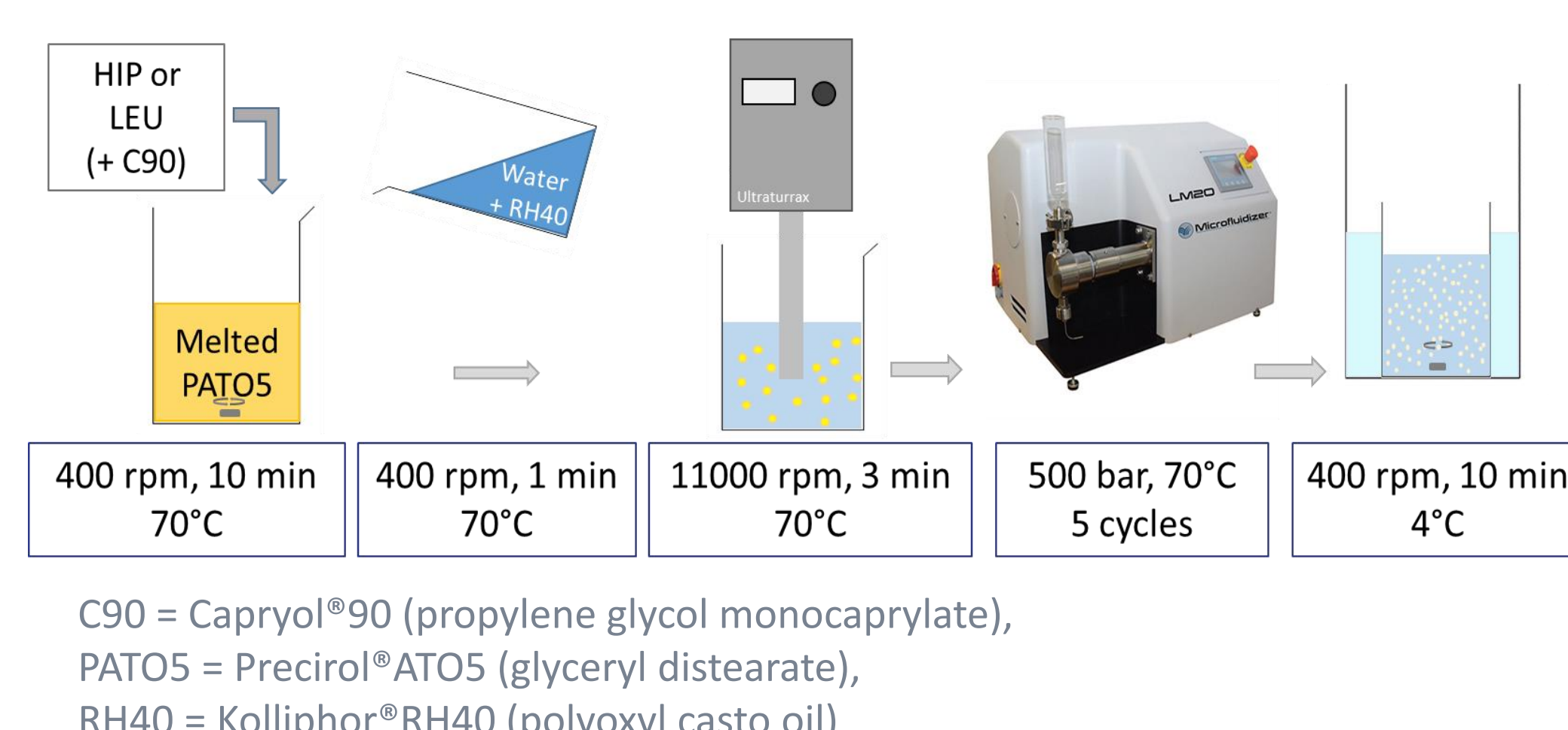


Fig. 1: Formation of a HIP by electrostatic interaction between Leuprolide and sodium docusate (molar ratio 1:2) [2]

$$\text{Precipitation Efficiency (PE, \%)} = 100 - \left(100 \times \frac{\text{Leuprolide concentration before HIP}}{\text{Leuprolide concentration after HIP}} \right)$$

NLC and SLN formulation by High Pressure Homogenization



C90 = Capryol®90 (propylene glycol monocaprylate),
PATO5 = Precirol®ATO5 (glyceryl distearate),
RH40 = Kolliphor®RH40 (polyoxyl casto oil)

Fig. 2: Set up for the preparation of loaded SLN via HPH method

$$\text{Encapsulation Efficiency (EE, \%)} = \frac{\text{Total amount of drug} - \text{unencapsulated amount of drug}}{\text{total amount of drug}} \times 100$$

Physico-chemical characterizations

Particle size was determined by Diffraction Light Scattering (DLS) and confirmed by Cryo-TEM measurements. PE and EE were measured by HPLC.

Enzymatic degradation by trypsin:

Trypsin solution (0.3 mg/mL) was added to samples ([LEU]=100 µg/mL) at 37°C. Enzymatic activity was stopped at predetermined time points. Remaining LEU was analyzed by HPLC.

Permeability evaluation across intestinal cell models:

SLN and NLC loaded with HIP and DiD (lipophilic dye) were used. Interactions with Caco-2 and Caco-2:HT29-MTX (3:1) monolayers were observed by **confocal microscopy (CLSM)**. Internalization by Caco-2 cells was evaluated by **flow cytometry**. Transport studies across Caco-2 and Caco-2/HT29-MTX monolayers were conducted in HBSS. Transported LEU was quantified via ELISA and the apparent permeability (P_{app}) was calculated :

$$P_{app} (\text{cm} \cdot \text{s}^{-1}) = \frac{\text{Transport rate}}{\text{Surface area} \times \text{initial LEU concentration}}$$

LEU release from SLN and NLC was studied in HBSS over 6 hours. LEU was quantified by HPLC.

RESULTS

Encapsulation efficiency

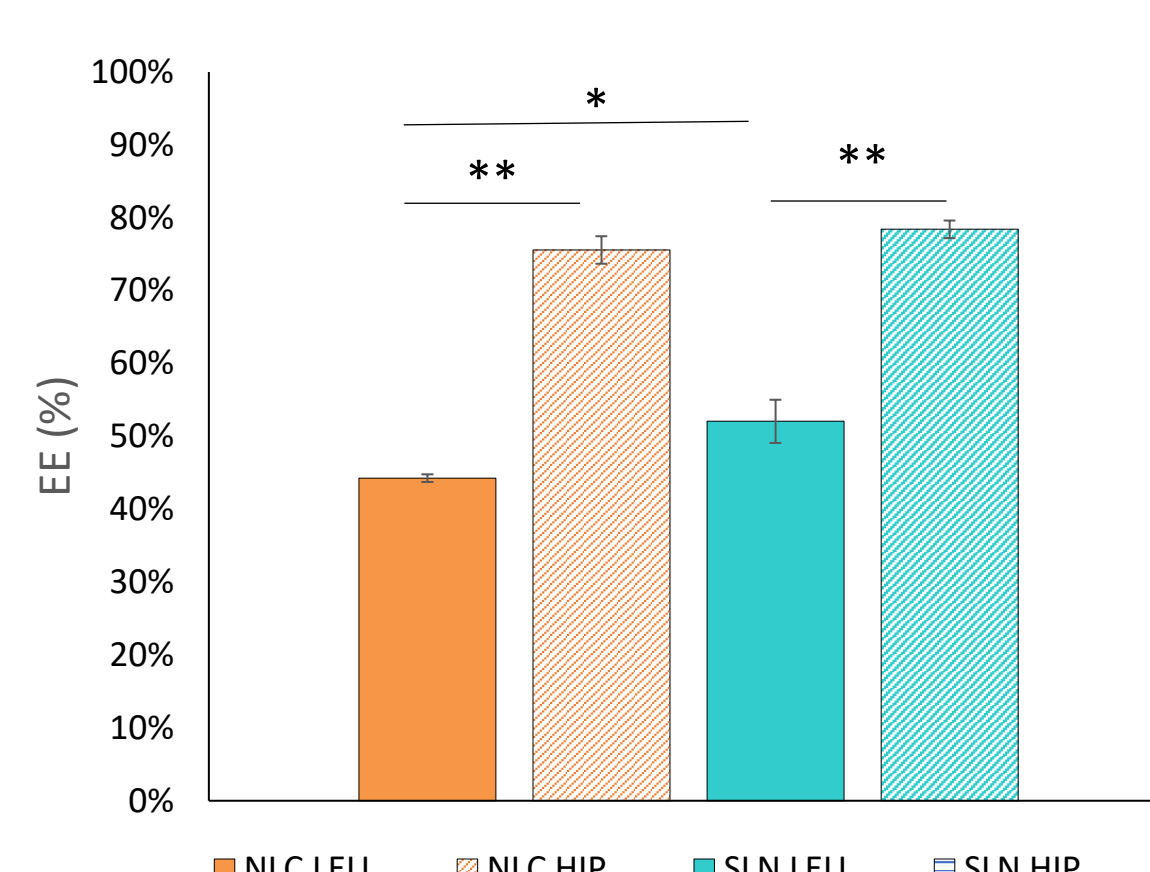


Fig. 3: HIP and LEU Encapsulation efficiency as a function of formulation type, (mean $\pm$ SEM * $p < 0.05$, ** $p < 0.01$) (n=3)

Precipitation efficiency was 99.9% (n=118).

Protection regarding trypsin activity

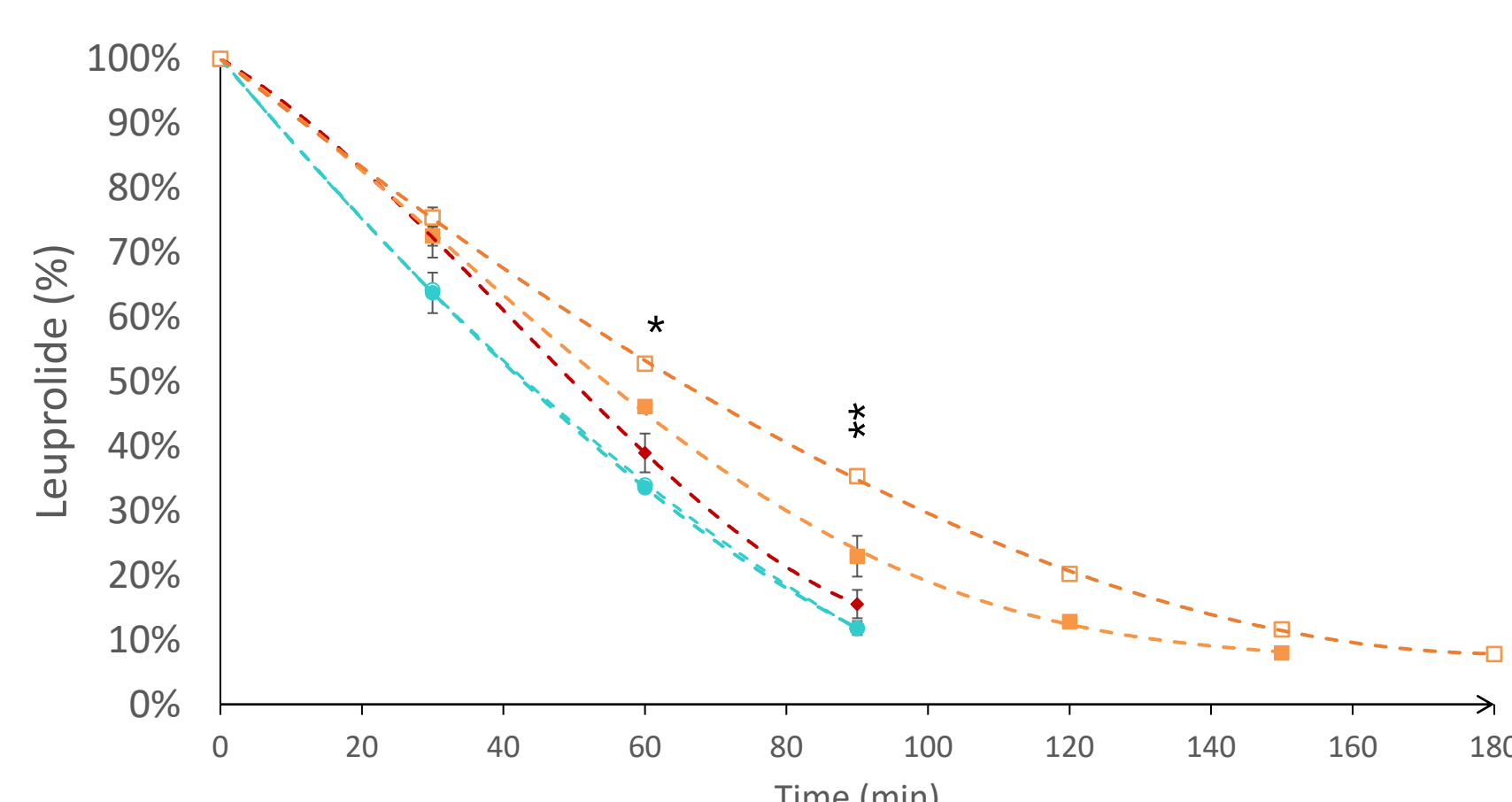


Fig 5: Degradation profile of LEU from NLC LEU (■), NLC HIP (□), SLN LEU (●) and SLN HIP (○) by trypsin, compared to free LEU (◆). Mean $\pm$ SEM (n=3) (* $p < 0.05$ and ** $p < 0.01$ when compared to suspensions dispersed in water)

Trypsin is not soluble in C90 [4].

Then, the addition of a C90 in NLC led to significant protection of LEU over trypsin degradation.

Physico-chemical characteristics

Table 1: Size distribution of NLC and SLN systems

	Blank NLC	NLC-LEU	NLC-HIP	Blank SLN	SLN-LEU	SLN-HIP
Z-average (nm)	114 $\pm$ 11	113 $\pm$ 1	125 $\pm$ 2	119 $\pm$ 4	124 $\pm$ 1	127 $\pm$ 1
PDI	0.2	0.2	0.2	0.2	0.2	0.2

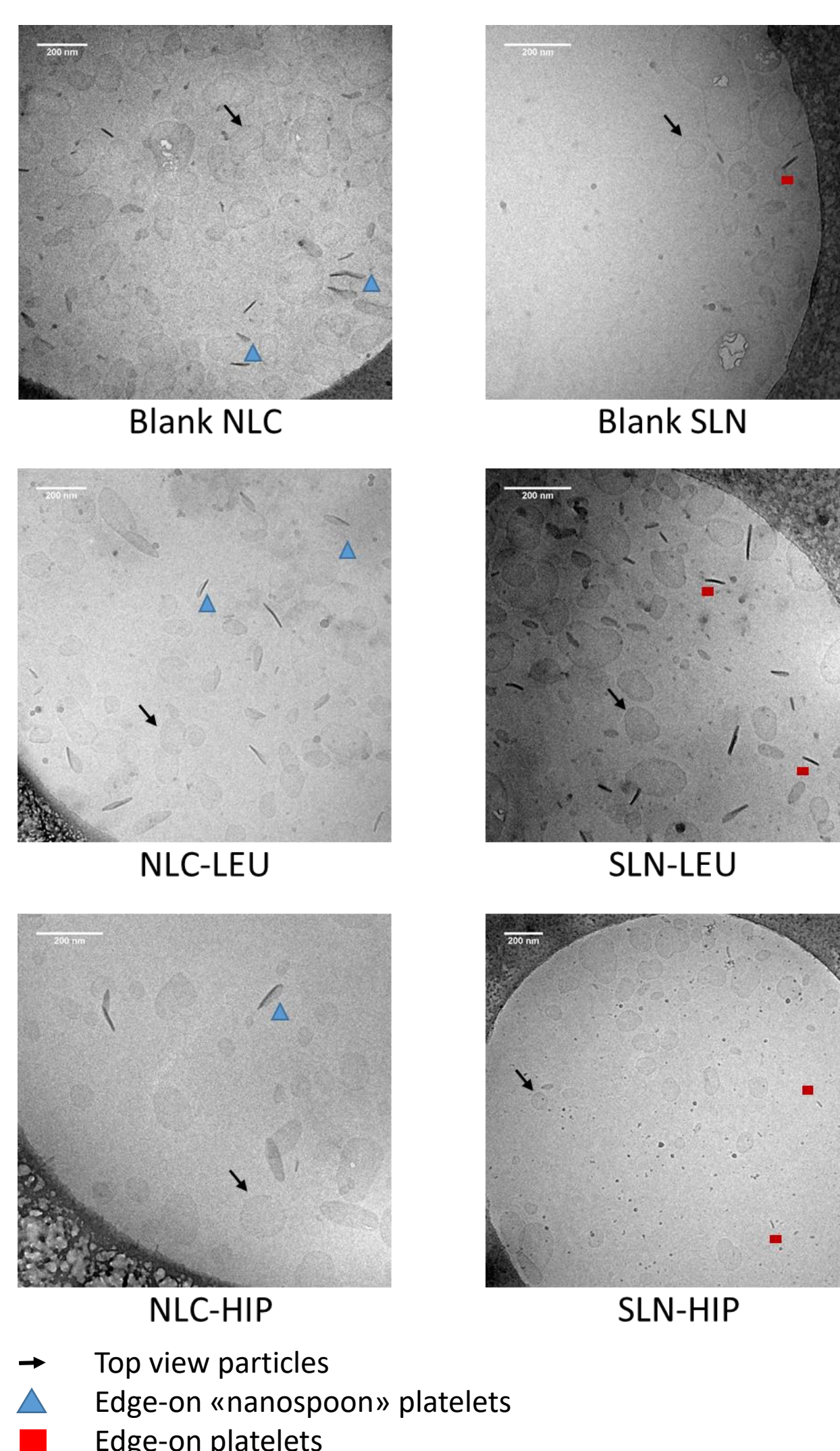


Fig. 4: Cryo-TEM images

Nanoparticles internalization by cell monolayers

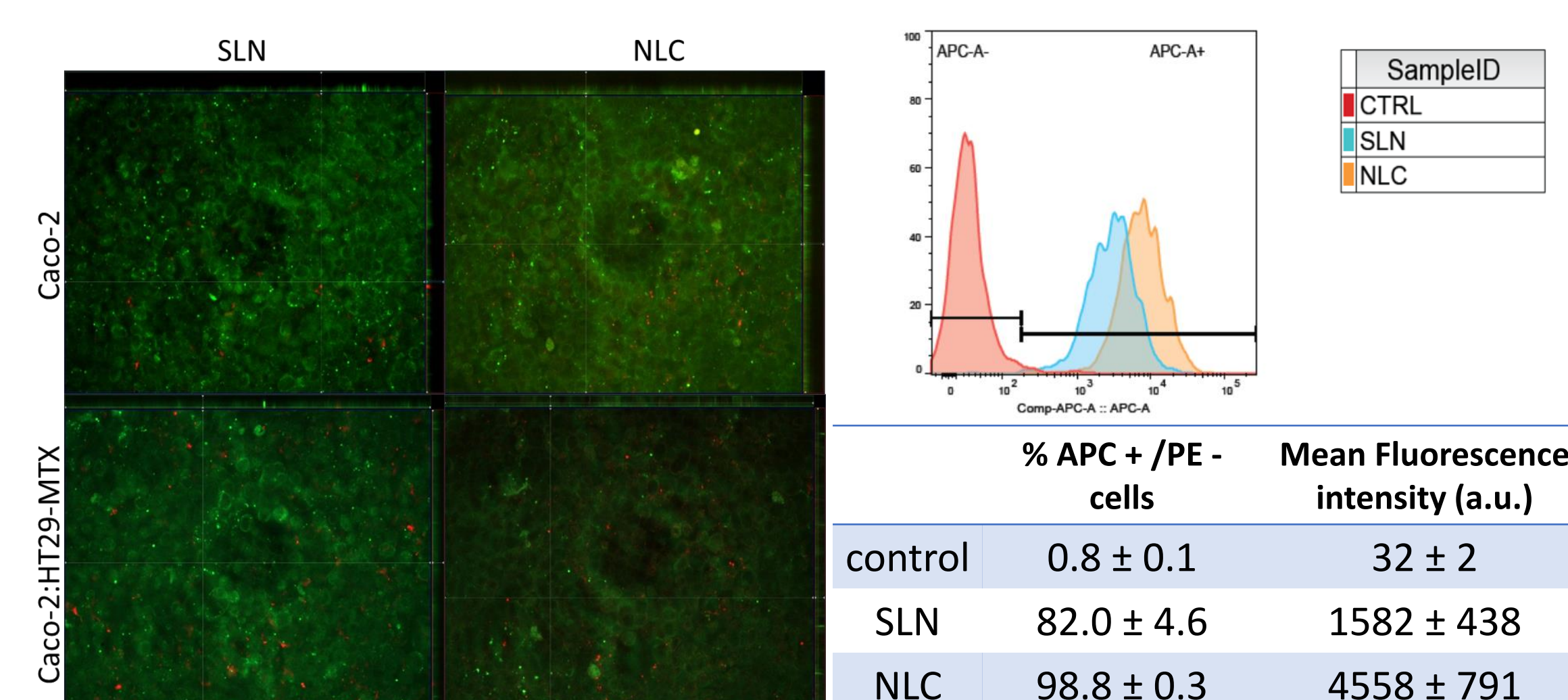


Fig. 6: (A) CLSM images of SLN-DiD and NLC-DiD (red channels) interactions with Caco-2 monolayers stained with rhodamine-phalloidine (green channel), x25

(B) Flow cytometry quantitative interaction study of Caco-2 cells (N=3, n=10, mean $\pm$ SEM)

Confocal microscopy images show SLN and NLC internalization by Caco-2 and Caco-2/HT29-MTX co-cultured cell monolayers, enlighting the ability of nanoparticles to cross mucus. Flow cytometry confirmed high particle uptake by Caco-2 [5].

Permeability evaluation

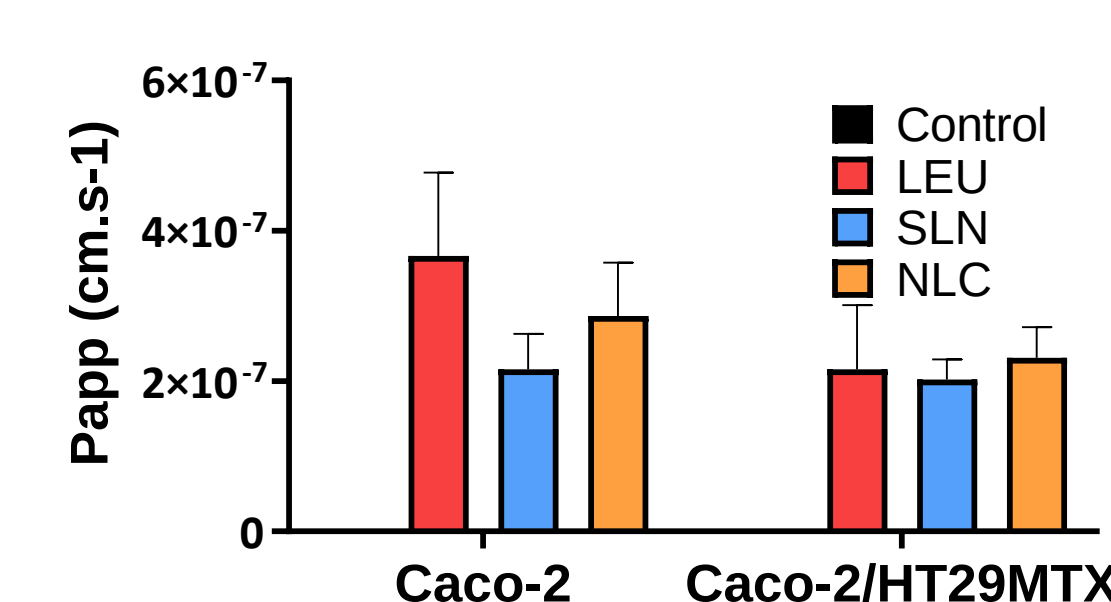


Fig7: P_{app} of neat and encapsulated LEU on Caco-2 and Caco-2/HT29-MTX cell monolayers (n=3, N=3, mean $\pm$ SEM)

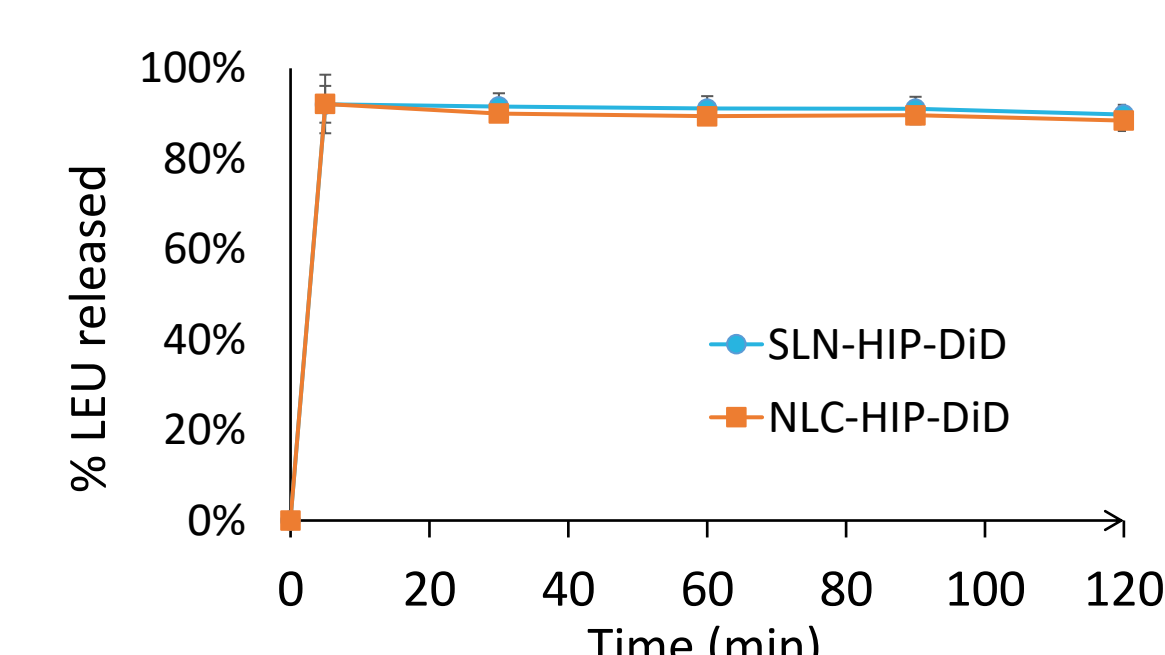


Fig. 8: In-vitro release profile of LEU in HBSS at 37°C (mean $\pm$ SEM, n=3)

The platelet-like structure of the particles, implying a large surface of exchange with the medium induced a high burst released of LEU from the nanoparticles [6]. Moreover, the HIP is not stable in this high ionic strength environment (150 mM) [7].

CONCLUSIONS

- Reproducible and monodisperse peptide-loaded SLN and NLC were obtained using HPH method, a **scalable and solvent free method**.
- Increasing peptide lipophilicity by HIP formation improved leuprolide encapsulation efficiency
- NLC provided a **significant protection** to trypsin-induced proteolytic degradation
- SLN and NLC were **internalized** by Caco-2 and Caco-2/HT29-MTX cell monolayers
- Important drug release in HBSS impeded permeability enhancement by encapsulation
- Stability of Leuprolide HIP needs to be improved to withstand biorelevant conditions and benefit from the advantages of lipid-based nanoparticles

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